

Unsteady Drag on Falling Coffee Filters Analysis

Humza Khan

University at Buffalo, Buffalo, NY, 14261, United States of America

Abstract

The main purpose of this laboratory experiment was to analyze the unsteady drag on falling coffee filters using two methods. The first method employed the use of a stopwatch to measure the fall time of coffee filter under 3 different scenarios (single filter, double filters, quadruple filters). The filters fell from a height of 2.96926 meters and the fall velocity was calculated using the time measured. This data was later processed to obtain C_D values. The second method involved the use of a high speed camera, and data obtained from the recorded videos (time, frame-rate) was used to solve for terminal velocities for each fall, which were then used to solve for C_D values from this approach. To draw a conclusion, the results from both methods were compared and their uncertainties were also analyzed. The C_D value from the high speed camera was lower than the C_D value from the whole height. Which means that when the coffee filters are accelerating, they are experiencing increased average drag when compared to when they are at their terminal velocity as they are in a steady state for a brief period until they stop on the ground.

I. Introduction

This lab involves using two different approaches to extrapolate the coefficient of drag (C_D) of the coffee filters. The first approach requires the time measurement of the full drop of the coffee filter in the drop column when it is in the unsteady state, averaging all the time values and solving for the drop velocity over the drop column's height.

Whereas, the second approach involves the data recorded via the high speed camera over the last few seconds of the drop. This data is crucial because this data is over the actual steady state where it is highly likely that the dropped filters have achieved a steady terminal velocity.

The main goal of the experiments performed was to calculate and compare the C_D from both of the methods. Since, we are assuming the coffee filters fall at a terminal velocity, the weight of the coffee filters will be in equilibrium with the drag force experienced by the filters. This relationship can be expressed via the equation below.

$$F_D = W = mg = \frac{1}{2} C_D \rho V_\infty^2 A_C \quad (1)$$

This equation can be rearranged such that the v^2 component is isolated on one side while everything else is on the other side. This way, we can write the v^2 in terms of m , g , ρ , C_D , and A_C .

$$m \left(\frac{2g}{C_D \rho A_C} \right) = V_{\infty}^2 \quad (2)$$

The velocities were obtained from the time that was measured for the first method, and the video data that was recorded from the high speed camera for the second method. The masses were known for each scenario. The equation above was plotted in the form of $Y = (\text{slope})X + C$ on Microsoft Excel, where the slope data was extrapolated using a visual plot and the LINEST() function. This extrapolated slope data was then used to solve for C_D for each method.

$$\frac{2g}{C_D \rho A_C} = \text{Slope} \rightarrow C_D = \frac{2g}{(\text{Slope}) \rho A_C} \quad (3)$$

Understanding the difference between both methods is important because the C_D obtained from the high speed camera was lower when compared to the C_D resulting from the data from the whole drop (unsteady state) of the first approach. This means that the overall drag experienced by an object in an unsteady state will be higher than if the same object was in steady state (in this case, terminal velocity).

II. Experimental Methods

The experiment was performed using coffee filters as due to their low weight and high drag properties. The filters were dropped in 3 different scenarios (1 filter, 2 filters, and 4 filters). The mass of the coffee filters was measured and averaged out to 0.68 grams. Furthermore, the coffee filters were assumed to be circular, so an average diameter was also calculated based on the innermost and the outermost points of the shape. This resulted in an average diameter of 0.15383333 meters. The average diameter, A_C (cross-sectional area), and the masses for each scenario are calculated in Tables 1, 2, and 3.

Diameter Data	
Outermost (m)	Innermost (m)
0.154	0.149
0.163	0.152
0.155	0.15
Innermost average	Outermost average
0.157333333	0.150333333
Average of both diameters	
0.153833333	

Table 1

# of Coffee Filters	total mass (kg)
1	0.00068
2	0.00136
4	0.00272

Table 2

Area of filter based on average diameter (m ²)
0.018586208

Table 3

Method 1 – Stopwatch Method Setup:

- The height of the transparent drop column was measured to be 116.9 inches using a measuring tape.
- A vacuum pump (figure 1) was used to transport the coffee filters to the top of the column.
- A guideline was used inside the column to ensure that the coffee filters dropped in a straight line.

- Once this was set up, the coffee filters were dropped stacked in 3 different scenarios which were either one, two, or four coffee filters were dropped at a time.
- Each scenario was repeated thrice.
- As the coffee filters were dropped, a team member was measuring the time for them using a phone stopwatch. This data was recorded by the team for processing the average drop velocity and then C_D .



Figure 1: Vacuum Pump

Method 2 – High Speed Camera Method Setup:

- The high speed camera (figure 2) was set up on the floor just outside the wall of the drop column.
- After the camera was set up, its position was not changed throughout the experiment and the fixed measurements L' and D' (figure 3) were recorded.
- The remaining measurements of L (figure 3) was recorded after each drop as it depended on where the coffee filters would land after the drop.
- The measurement for D (figure 3) was later processed using similar triangle method.
- One of the team members was operating the high speed camera using the 'Ueye Cockpit' software and recording appropriate videos for each drop. (Maximum frame rates were set to 4096)
- The videos were exported in the most accessible format to UBBox (online cloud storage) where all the team members could access it on their own time.
- The data from the videos was access through software like VLC Media Player and was used to solve for the terminal velocity and then eventually C_D .

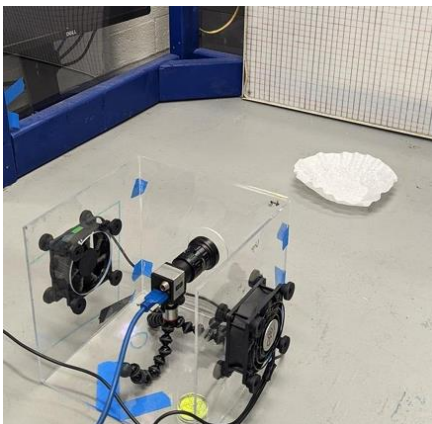


Figure 2: High Speed Camera

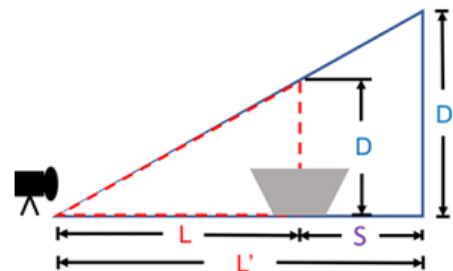


Figure 3: Camera Geometry

Note: Experiments for both methods 1 and 2 were being performed simultaneously. The time and video data were being recorded by one dedicated member each. The rest of the team was conducting the experiment. This was done to keep any human errors such as reaction times consistent and not random, ensuring that the average takes them into account and they are mitigated.

III. Data Analysis

The following steps were followed for the data analysis of each of the methods:

Method 1 – Stopwatch Method Data Analysis:

1. The time for each scenario was recorded 3 times and its average was calculated and tabulated in table 4.

1 Filter Drop time average			
trial #1 time (s)	trial #2 time (s)	trial #3 time (s)	Average time (s)
3.33	3.75	3.66	3.58

2 Filters Drop time average			
trial #1 time (s)	trial #2 time (s)	trial #3 time (s)	Average time (s)
2.78	2.5	2.43	2.57

4 Filters Drop time average			
trial #1 time (s)	trial #2 time (s)	trial #3 time (s)	Average time (s)
2	2.25	2.3	2.183333333

Table 4

2. The average times were used to solve for the average drop velocity for each case over the drop column's height, which was 2.969265939 meters when converted from inches. The resulting velocity for each case is tabulated in table 5.

Average velocities (m/s)	
1 filter	0.829403893
2 filters	1.155356396
4 filters	1.359969132

Table 5

3. The obtained average velocities were then squared and tabulated with their respective scenario's mass. The data is displayed in table 6. Doing this will allow us to have an equation in form of equation (2), where the slope data can be used to solve for C_D .

Mass of coffee filters (kg)	$V_{avg}^2 (m^2/s^2)$
0.00068	0.687910818
0.00136	1.334848403
0.00272	1.84951604

Table 6

4. The data in table 6 was used to plot figure 4.

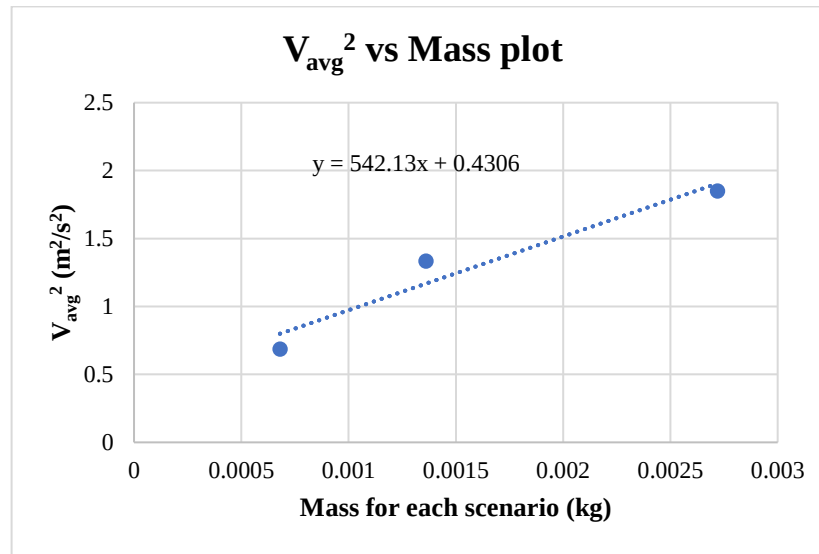


Figure 4

5. The data from table 6 was also used as an input for the LINEST() function on Excel, which resulted in the slope data tabulated in table 7.

Slope	542.1311477	0.430577	y-intercept
Slope Uncertainty	141.7675454	0.255055535	y-intercept uncertainty

Table 7

6. Recalling equation (3), we can plug in the slope value obtained from the LINEST() function in it and along with the rest of the known values, resulting in C_D .

$$C_D = \frac{2g}{(\text{Slope})\rho A_c} = \frac{2(9.81)}{(542.1311)(1.225)(0.0186)} = 1.589526496$$

7. Uncertainty of C_D was calculated to be about **0.416175303**. (Please note that the uncertainty calculation is provided in the Appendix as it was hand calculated)
8. Therefore, the resulting C_D along with its uncertainty for method 1 is:

$$C_D \pm \delta C_D = 1.589526496 \pm 0.416175303$$

Method 2 – High Speed Camera Method Data Analysis:

1. For each drop, D was calculated using the similar triangle rule as the other values like L, L', and D' were already obtained during the experiment. Time was extracted from the videos after running them frame by frame. There was add-on available on VLC Media Player which allowed us get the exact time values as each frame along with the (time/frame) value. V_{term} and V_{term}^2 were solved based on the total time and the D measurement calculated. All the data is tabulated in table 8. The fractional data used to solve D is recorded in table 9.

Information	Start time (s)	End time (s)	Total time (s)	D' (cm)	D (cm)	D (m)	V_{term}	V_{term}^2	s/frame
1 filter trial #1	3.624	3.962	0.338	54.5	23.22345395	0.232234539	0.687084436	0.472085023	0.003248008
1 filter trial #2	3.278	3.681	0.403	63	26.84546053	0.268454605	0.66614046	0.443743112	0.003353004
1 filter trial #2	5.165	5.611	0.446	48	20.45368421	0.204536842	0.458602785	0.210316514	0.003258008
2 filter trial #1	2.304	2.519	0.215	50.9	21.68942763	0.216894276	1.008810588	1.017698801	0.003233003
2 filter trial #2	2.162	2.411	0.249	58.4	24.88531579	0.248853158	0.999410273	0.998820893	0.003233003
2 filter trial #3	2.051	2.273	0.222	53.3	22.71211184	0.227121118	1.023068101	1.046668339	0.003326005
4 filter trial #1	2.394	2.596	0.202	61.3	26.12105921	0.261210592	1.293121743	1.672163842	0.003306004
4 filter trial #2	2.332	2.527	0.195	58.6	24.97053947	0.249705395	1.280540486	1.639783936	0.003263005
4 filter trial #3	2.368	2.556	0.188	59.5	25.35404605	0.253540461	1.348619471	1.818774477	0.003293005

Table 8

D'(cm)	32.385
L'(cm)	76
D'/L'	0.426118

Table 9

2. Similar to what was done for Method 1, the squared terminal velocity values were tabulated along with their respective experimental masses so that they can be plotted. The values are recorded in table 10.

Mass (kg)	$V_{\text{term}}^2 \text{ (m}^2/\text{s}^2)$
0.00068	0.472085023
0.00068	0.443743112
0.00068	0.210316514
0.00136	1.017698801
0.00136	0.998820893
0.00136	1.046668339
0.00272	1.672163842
0.00272	1.639783936
0.00272	1.818774477

Table 10

- The values from table 10 are plotted. V_{term} is squared in accordance with equation (2)'s y-data, and mass being the x-data. The plot is represented in figure 5.

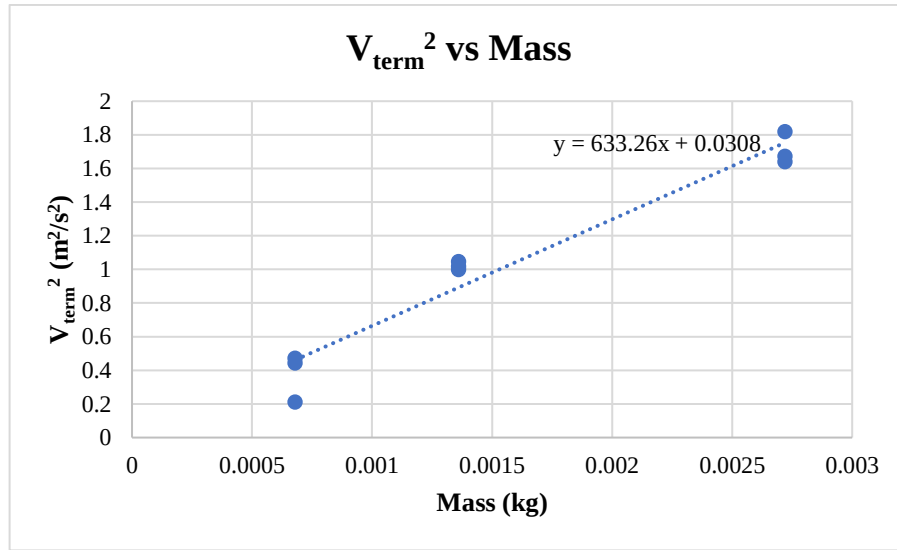


Figure 5

- The data from table 10 was also used as the input for the LINEST() function in order to obtain the slope data and its uncertainty. The results from the LINEST() function are tabulated in table 11.

Slope	633.2579	0.030793	y-intercept
Slope Uncertainty	55.25092	0.099403	y-intercept uncertainty

Table 11

- Again, we will use the slope data and the slope uncertainty and input them in equation (3) to obtain C_D from the second method.

$$C_D = \frac{2g}{(\text{slope})\rho A_c} = \frac{2(9.81)}{(633.2579)(1.225)(0.0186)} = 1.360791$$

- Uncertainty of C_D was calculated to be about 0.12038151648. (Please note that the uncertainty calculation is provided in the Appendix as it was hand calculated)
- Therefore, the resulting C_D along with its uncertainty for method 2 is:

$$C_D \pm \delta C_D = 1.360791 \pm 0.12038151648$$

IV. Results & Discussions

After analyzing all the data from both approaches, it can be said with certainty that the results for the 'High Speed Camera' method are more accurate. The uncertainty values for this method are much smaller when compared to the uncertainty values of the 'stopwatch' method. Furthermore, the high speed camera method captured data when the coffee filters were in steady state, which in comparison, has less discrepancies of the averaged out data from the stopwatch method where the coffee filters were in an unsteady state until they reached at the end of the drop column.

The provided C_D value of 0.644 does not come close to the obtained C_D values from both of the methods, even when the error margins are considered. This is likely because of the experiment being performed in a vastly different environment. Furthermore, the wind tunnel C_D value was obtained with rigid filter paper, which can be placed in the higher windspeeds without losing its shape. The experiment that was performed for this lab had non-rigid filter papers that would lose their original shape under small windspeeds, which can drastically affect the C_D values.

The sources of error in the calculations can exist from multiple sources. For example, the tape measure used to measure the height of the drop column had an uncertainty of about $1/16^{\text{th}}$ of an inch. Similarly, the rulers used measure values like the diameters of the coffee filter, L' , D' , and L had an uncertainty of 0.1 cm. Furthermore, the reaction time from the team members who were operating the camera and the stopwatch definitely added to the uncertainty. Sources like these can add to the uncertainties in our calculations.

V. Conclusion

It can be conclusively deduced that the two methods represent a real life scenario (stopwatch method) and an ideal scenario (high speed camera method). The C_D in the real life scenario will always change at each timestep until terminal velocity is reached. However, the C_D will remain steady after the terminal velocity is reached as was calculated in the high speed camera method. Additionally, it can be deduced that the data obtained from a steady state scenario is much more precise as compared to the data from an unsteady scenario averaged out. This can be observed just by comparing the calculated C_D values for either methods. An improvement that can be made for this experiment would be using a more rigid body than coffee filters. They are prone to changing shape due to normal handling and this can affect the C_D values a lot.

VI. Appendix

Uncertainty Hand Calculations (Screenshots from the tablet):

C_D Equation ①

$$C_D = \frac{2g}{\rho A_c s}$$

$s = \text{slope.}$
slope value from
linear function

② $\frac{\partial C_D}{\partial s} = \frac{2g}{\rho A_c} (s^{-2}) = \frac{-1(2g)(s^{-2})}{\rho A_c}$

$$= \boxed{\frac{-2g}{\rho A_c s^2}}$$

③ $\partial(s) = \text{from Linear function}$

④ $\frac{\partial C_D}{\partial A_c} = -\frac{2g}{\rho s (A_c)^2}$

⑤ $A = \frac{\pi D^2}{4}$ ⑥ $\partial d = 0.001 \text{ m.}$

$$\frac{\partial A}{\partial D} = \boxed{\frac{\pi D}{2}}$$

$$\textcircled{1} \quad \delta A = \sqrt{\left(\frac{dA}{\partial D}\right)^2 (\partial D)^2} \quad \textcircled{2}$$

$$= \sqrt{\left(\frac{\pi^2 (0.1538333)^2}{4}\right) \cdot (0.001)^2}$$

$$\boxed{\delta A = 0.00024164083 \text{ m}^2}$$

Now, solving for scenario with stop watches. \textcircled{3}

$$\textcircled{*} \quad \frac{\partial C_D}{\partial S} = -\frac{2g}{\pi A_c S^2} = \frac{-2 \times 9.81}{1.225 \times (0.018586208) \times (542.1311477)^2}$$

$$= \boxed{-0.00293199618}$$

$$\textcircled{*} \quad \partial(S) = \text{from LINEST function}$$

$$\boxed{S(S) = 141.7675459}$$

$$\textcircled{*} \quad \frac{\partial C_D}{\partial A_c} = \frac{-2g}{\pi S(A_c)^2} = \frac{-2 \times 9.81}{1.225 \times (542.1311477) \times (0.018586208)^2}$$

$$= \boxed{-85.52182662}$$

$$\textcircled{*} \quad \boxed{\delta A = 0.00024164083 \text{ m}^2}$$

$$\delta C_D = \sqrt{\left(\frac{\partial C_D}{\partial s}\right)^2 (\delta(s))^2 + \left(\frac{\partial C_D}{\partial A_c}\right)^2 (\delta A)^2}$$

$$\delta C_D = 0.416175303$$

$$C_D \pm \delta C_D = 1.589526496 \pm 0.416175303$$

Now, for the Second Scenario

$$\begin{aligned} \textcircled{*} \frac{\partial C_D}{\partial s} &= -\frac{2g}{\pi A_c s^2} = \frac{-2 \times 9.81}{1.225 \times (0.018586208) \times (633.2579)^2} \\ &= -0.00214887368756, \end{aligned}$$

$\textcircled{*} \delta(s) = \text{from LINEST function}$

$$\delta(s) = 55.25092.$$

$$\begin{aligned} \textcircled{*} \frac{\partial C_D}{\partial A_c} &= \frac{-2g}{\pi s (A_c)^2} = \frac{-2 \times 9.81}{1.225 \times (633.2579) \times (0.018586208)^2} \\ &= -79.2151087525 \end{aligned}$$

$$\textcircled{*} \delta A = 0.00024164083$$

$$\delta C_D = \sqrt{\left(\frac{\partial C_D}{\partial s}\right)^2 (\delta s)^2 + \left(\frac{\partial C_D}{\partial A_c}\right)^2 (\delta A)^2}$$

6

$$\delta C_D = 0.12038151648$$

$$C_D + \delta C_D = 1.360791 \pm 0.12038151648,$$